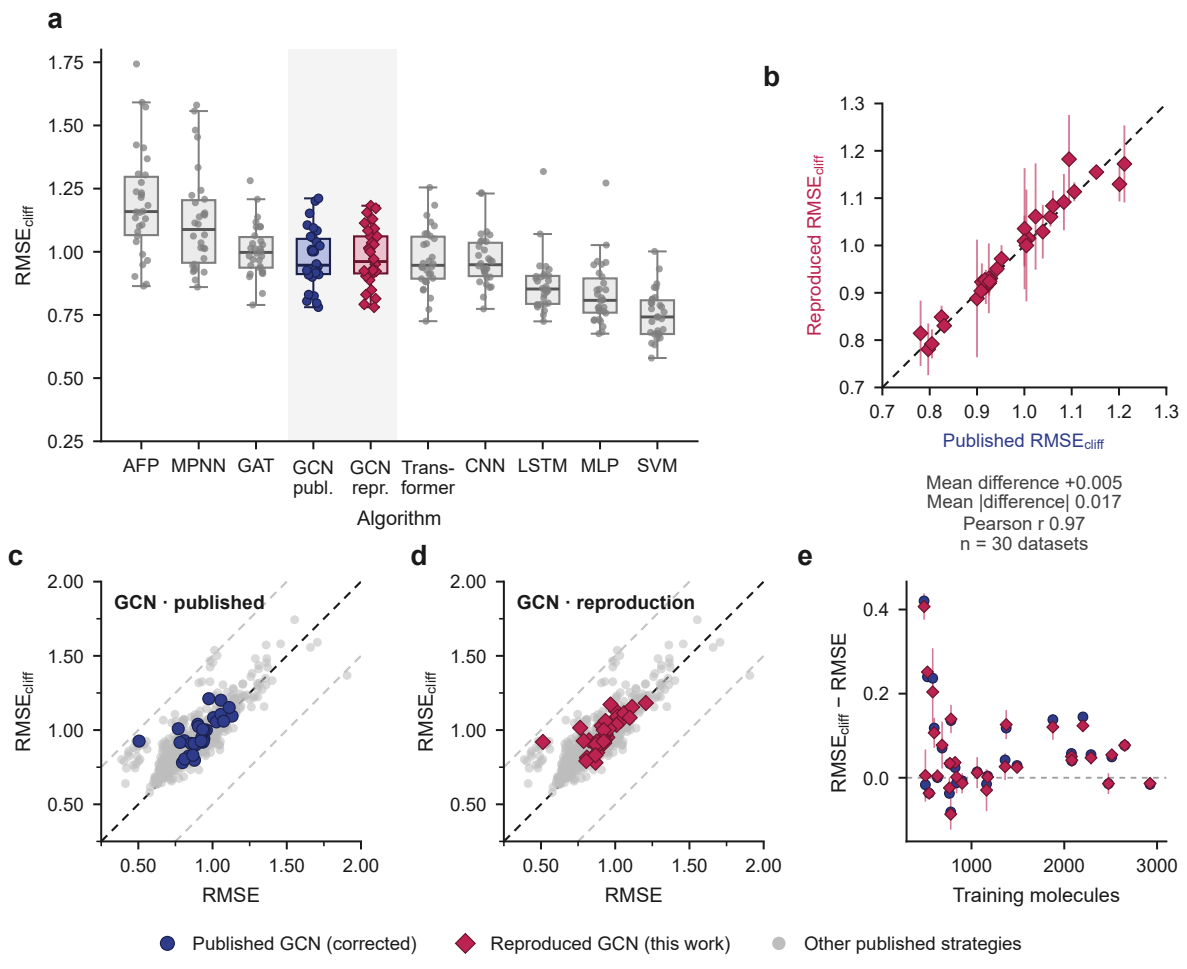

a
RMSE_cliff
1.75
1.50
1.25
1.00
0.75
0.50
0.25
AFP
MPNN
GAT
GCN publ.
GCN repr.
Trans-former
CNN
LSTM
MLP
SVM
Algorithm
b
Reproduced RMSE_cliff
Published RMSE_cliff
1.3
1.2
1.1
1.0
0.9
0.8
0.7
Mean difference +0.005
Mean |difference| 0.017
Pearson r 0.97
n = 30 datasets
c
GCN · published
RMSE_cliff
RMSE
2.00
1.50
1.00
0.50
d
GCN · reproduction
RMSE_cliff
RMSE
e
RMSE_cliff − RMSE
0.4
0.2
0.0
1000
2000
3000
Training molecules
Published GCN (corrected)
Reproduced GCN (this work)
Other published strategies